ASTROPHYSICS LABORATORY
Observational Astrophysics 2025

I) Basic notions in Observational astrophysics

a) Properties of the astronomical objects

- **Magnitude** Absolute magnitude
  Magnitude and distance: ly, parsec
- **Temperature and colour**
- **Size**

b) Preparing astronomical observations

- Visibility of the target: coordinates and time
- Telescope
- Detector and data acquistion

II) Observations: data acquisition

III) Data reduction and analysis

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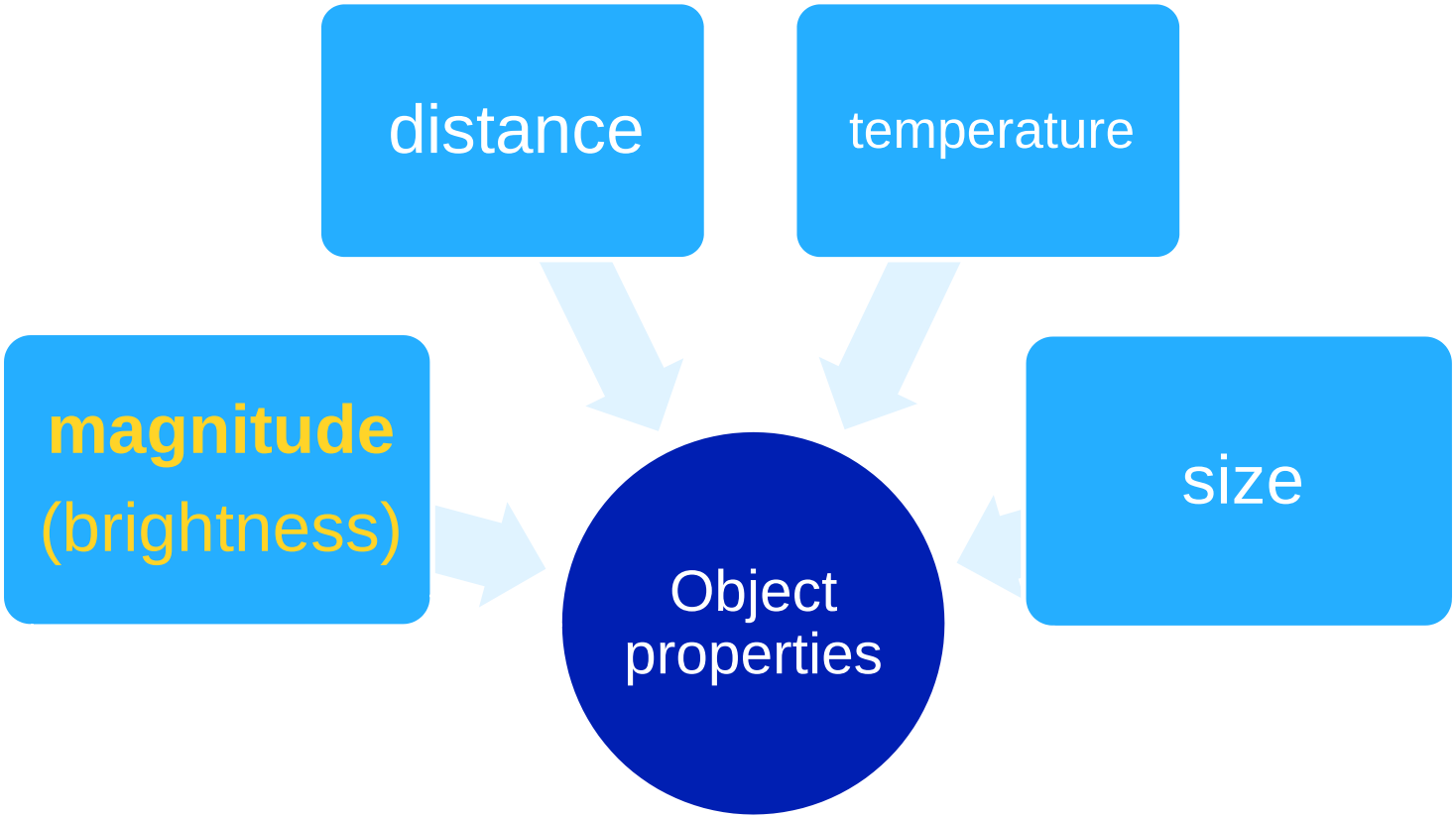
- Visibility of the target: coordinates and time
- Telescope
- Detector and data acquistion

II) Observations: data acquisition

III) Data reduction and analysis

Lecture notes for the LA@IST by A. M. Mourão, J. D. Duarte, S.Gonzalez-Gaitan, R. P. Santos

PROPERTIES OF THE TARGETS



Brightness

Magnitude



Brightness

intrinsic

magnitude

apparent

observer on Earth

absolute

observer at D=10 pc

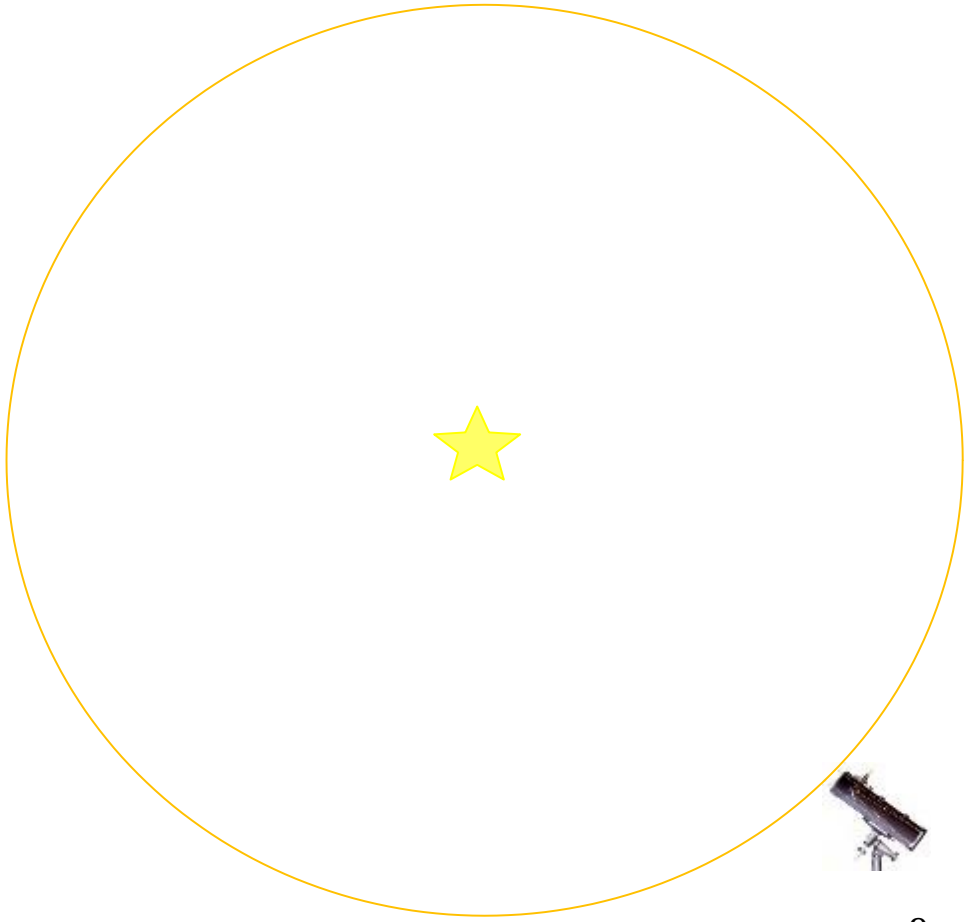
Brightness of the object

Magnitude



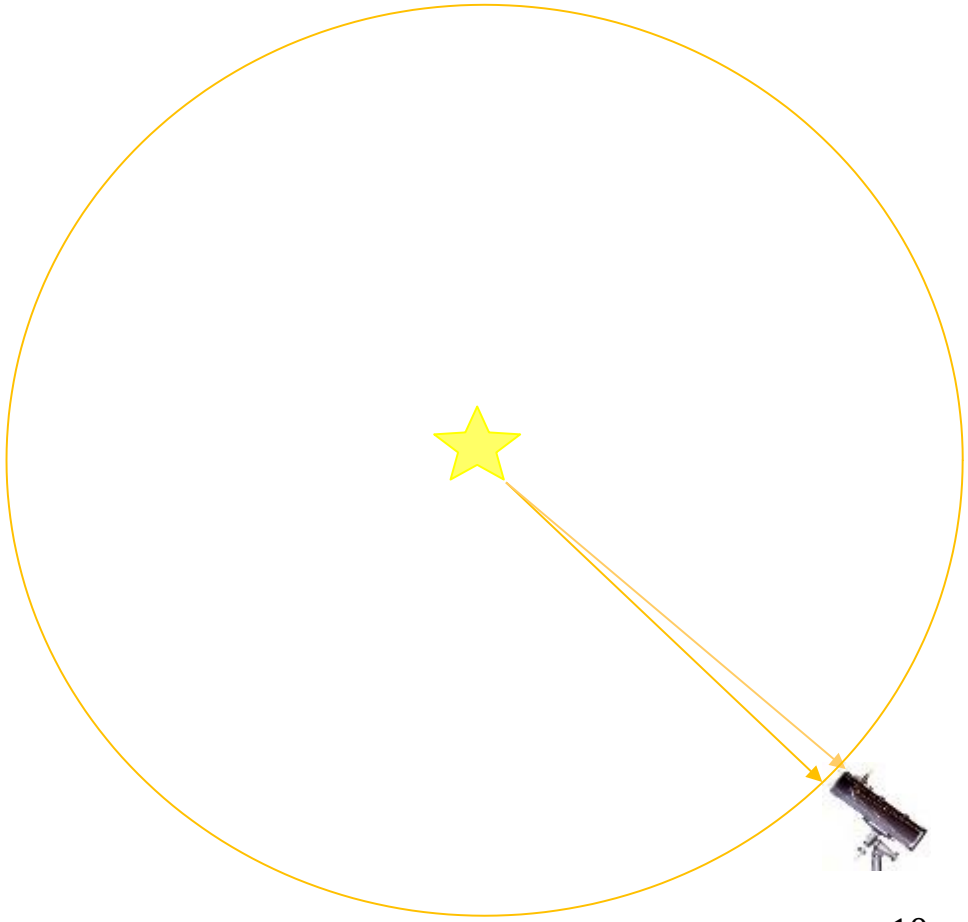
Brightness of the object

Magnitude



Brightness of the object

Magnitude



Brightness of the object
Magnitude

Flux **F:** ***energy/time/area***

Units: **W/m²**

erg/s/cm²

$$mag\ A = -2{,}5 \log F_A + Constant$$



Brightness of the object
Magnitude

Flux **F:** ***energy/time/area***

Units: **W/m²**

erg/s/cm²

$$mag\ A = -2{,}5 \log F_A + Constant$$

$$mag\ A - mag\ B = -2{,}5 \left(\log F_A - \log F_B \right)$$

$$-2{,}5 \left(\log \frac{F_A}{F_B} \right)$$



Brightness of the object
Magnitude

$$mag\ A = -2{,}5 \log F_A + Constant$$

$$mag\ Vega = -2{,}5 \log F_{Vega} + Constant$$

Definition

$$mag\ Vega = 0 = -2{,}5 \left(\log F_A \right) + Constant$$



$$2{,}5 \log F_{Vega} = Constant$$



Brightness of the object
Magnitude

$$m_A - m_B = -2{,}5\left(\log F_A - \log F_B\right) = -2.5 \log\left(\frac{F_A}{F_B}\right)$$

Example 1:

Let's consider two stars, A and B

$$m_A - m_B = 1$$

$$1 = -2.5 \log\left(\frac{F_A}{F_B}\right) \Rightarrow \frac{F_B}{F_A} = 10^{1/2.5} \simeq 2.511$$

B is brighter!!!

Brightness of the object
Magnitude

Example 2:

Let's consider two stars,

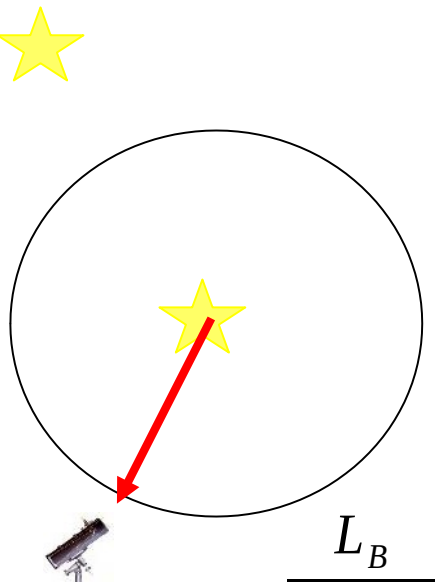
A and B stars:

$$m_A - m_B = 5 \rightarrow F_B / F_A = 100$$

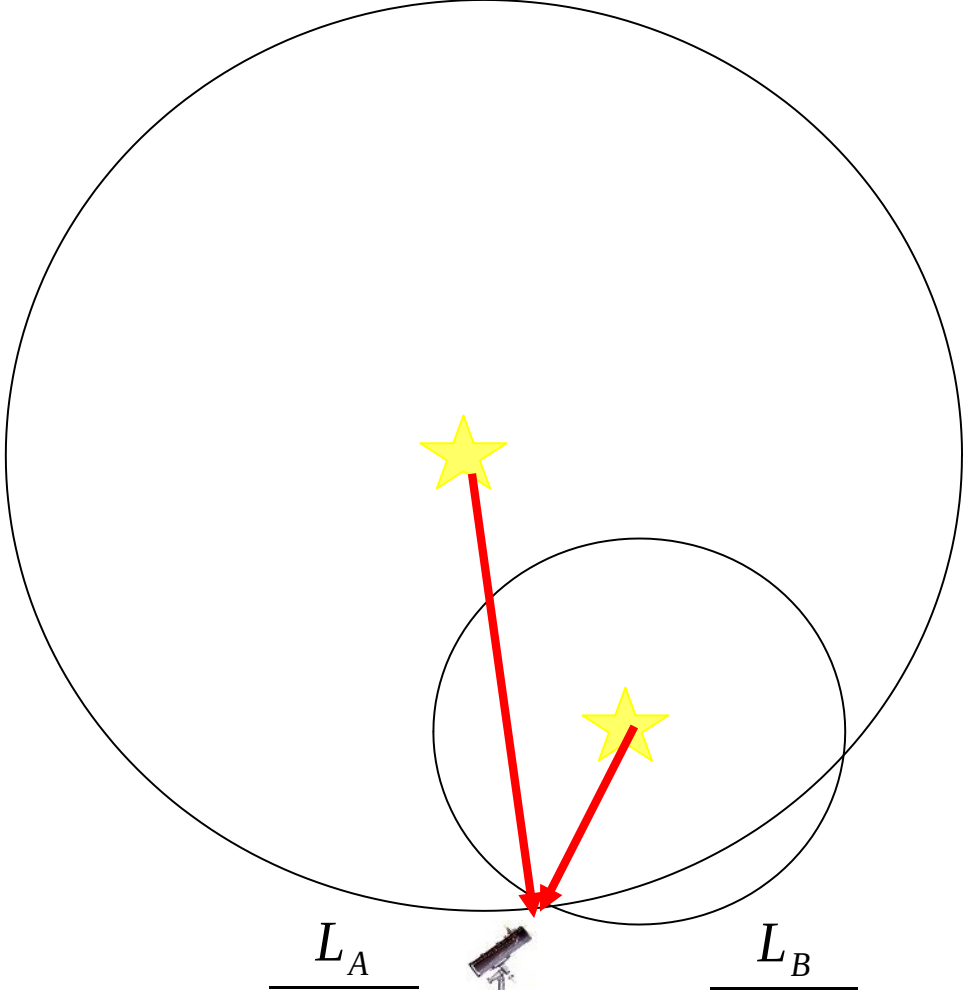


Hipparchus, 190 BC-120 BC

Apparent magnitude and distance



$$\frac{L_B}{4\pi D_B^2}$$



$$\frac{L_A}{4\pi D_A^2}$$

$$\frac{L_B}{4\pi D_B^2}$$

$$F_A$$

$$F_B$$

Apparent magnitude and distance

Two equal stars (A e B) at distances D_A and D_B to the Earth

What is the diference between the apparent magnitudes?

$$F_A = \frac{L_A}{4\pi D_A^2} \qquad F_B = \frac{L_B}{4\pi D_B^2}$$

$$m_A - m_B = -2{,}5 \log\left(D_B^2 / D_A^2\right) = 5 \log\left(D_A / D_B\right)$$

Consider: D_B=10 pc

definition: M=mag B ***Absolute Magnitude M: mag @10pc***

$$\mu = \underbrace{m_A - M} = 5 \log\left(D_A / 10\right)$$

D_A-Luminosity distance

μ- Distance modulus

Apparent magnitude and distance

Two *equal stars* A e B at distances D_A and D_B to the Earth

What is the diference between the apparent magnitudes?

$$F_A = \frac{L_A}{4\pi D_A^2} \qquad F_B = \frac{L_B}{4\pi D_B^2}$$

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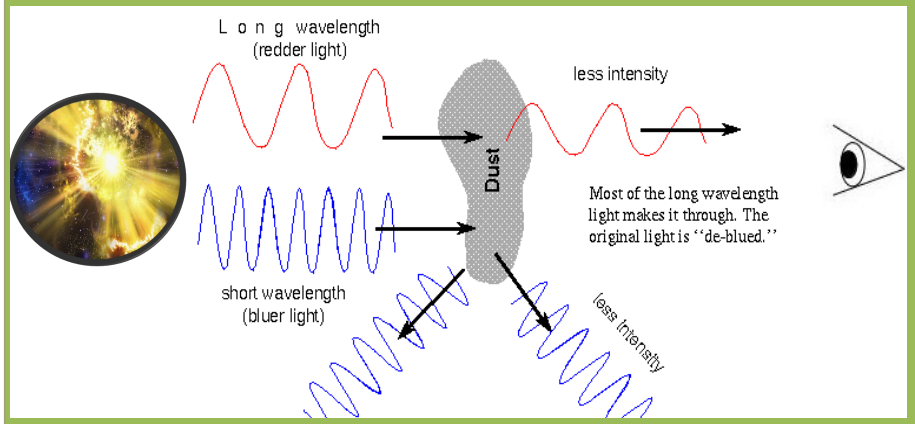
Source of errors:

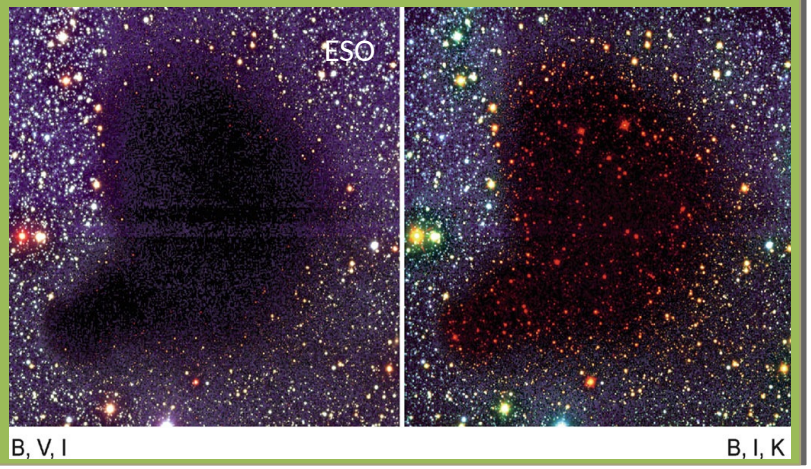
extinction of light due to dust

Apparent magnitude and distance

Extinction: absorption of light by dust

Reddening: reemission at longer wavelengths

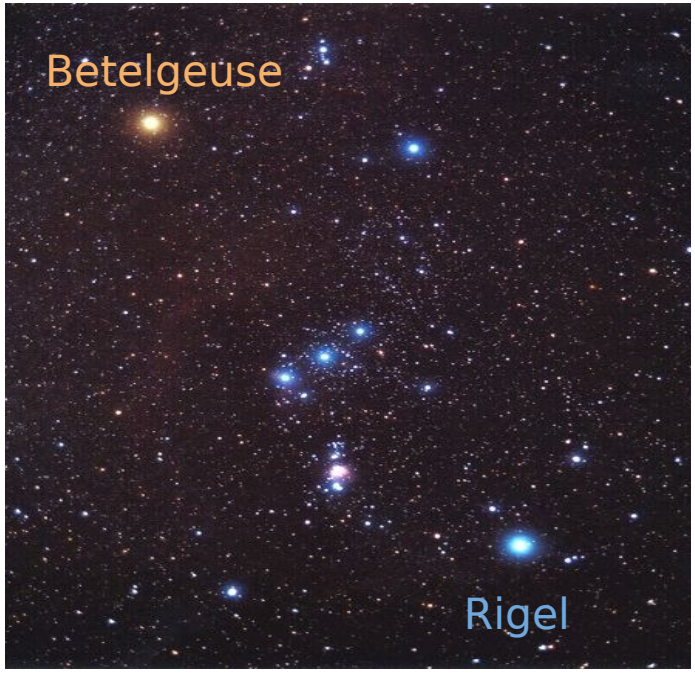




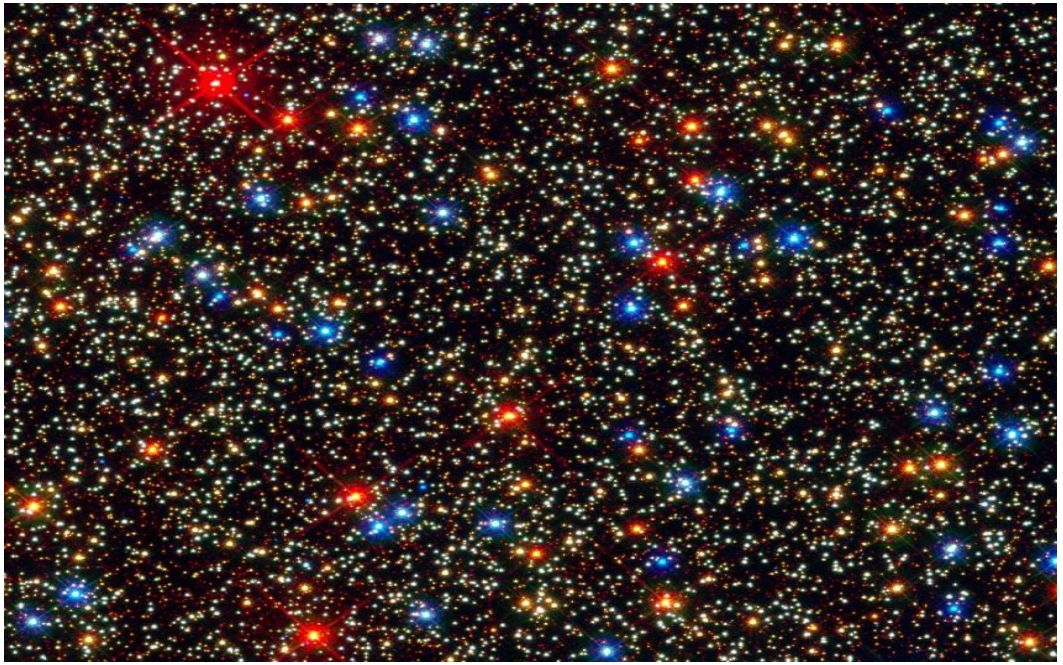
Distance Modulus corrected for the extinction:

$$\mu = m - M = 5\log\left(D_A/10\right) + A$$

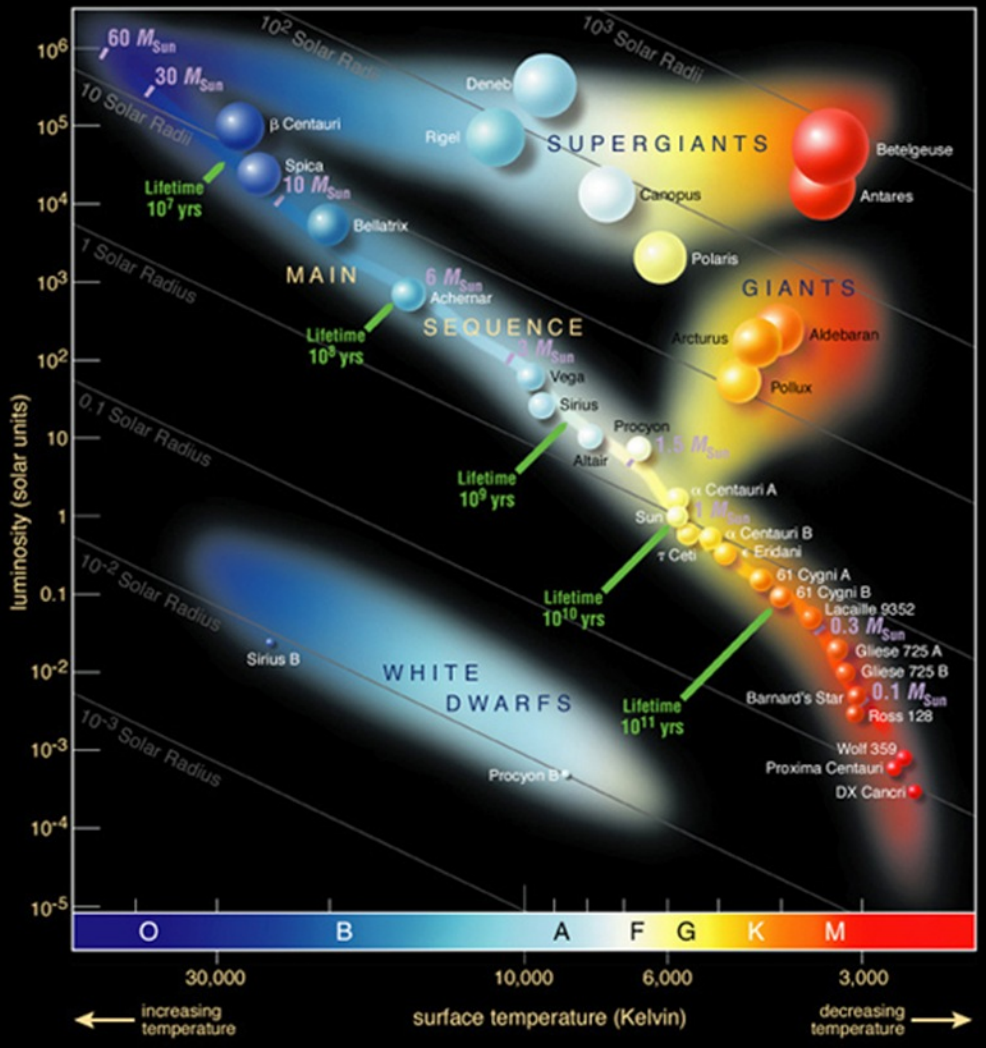
Apparent magnitude and colours



Orion Constellation



Central part of the Omega Cen globular cluster (HST)



Deneb

Arcturus

Aldebaran

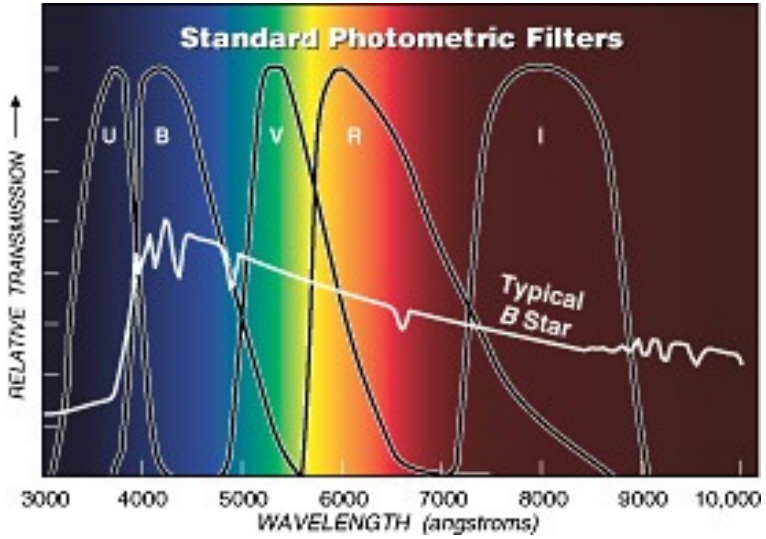
Vega

Altair

Sun

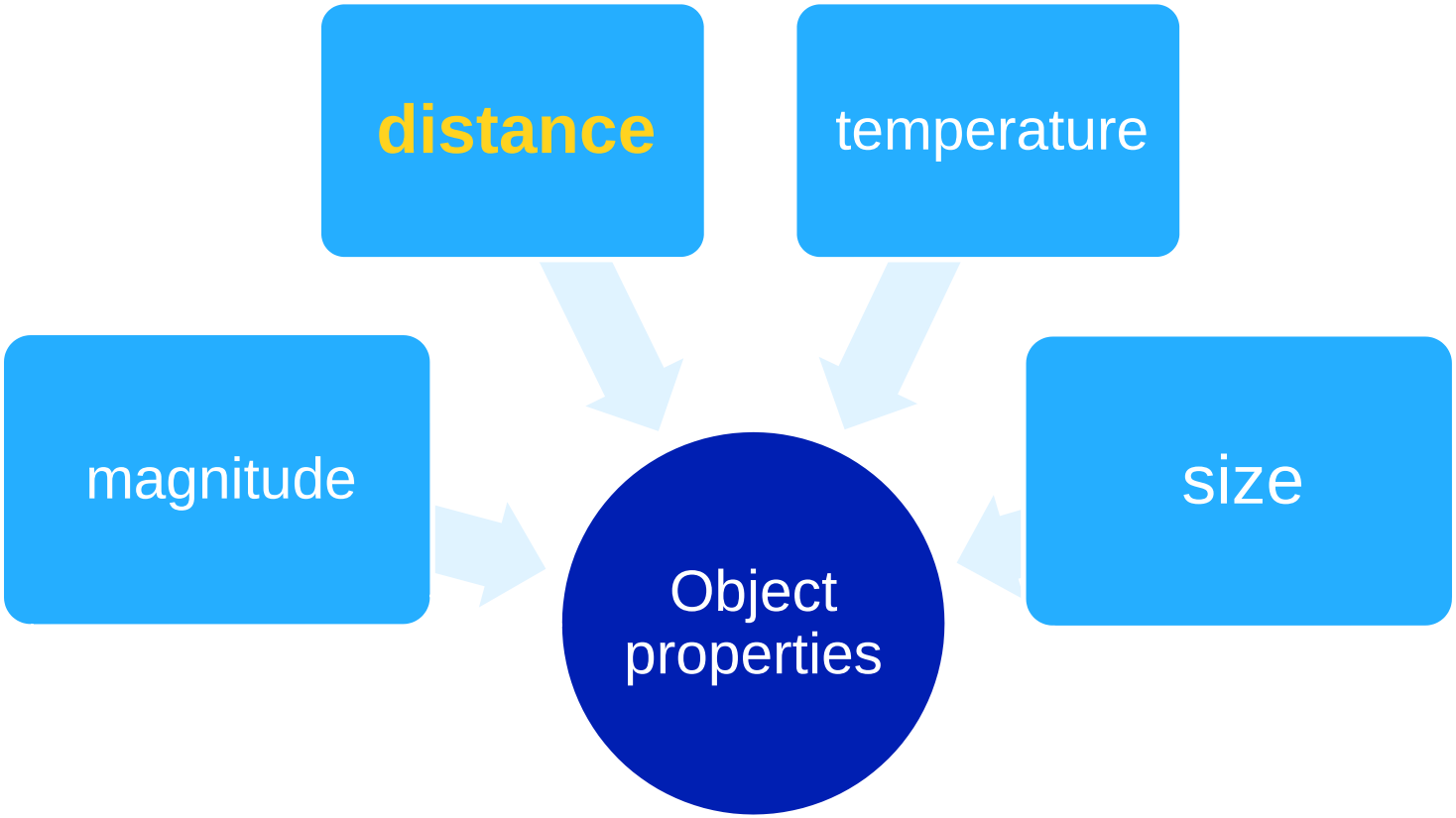
Sirius B

Apparent magnitude and colours

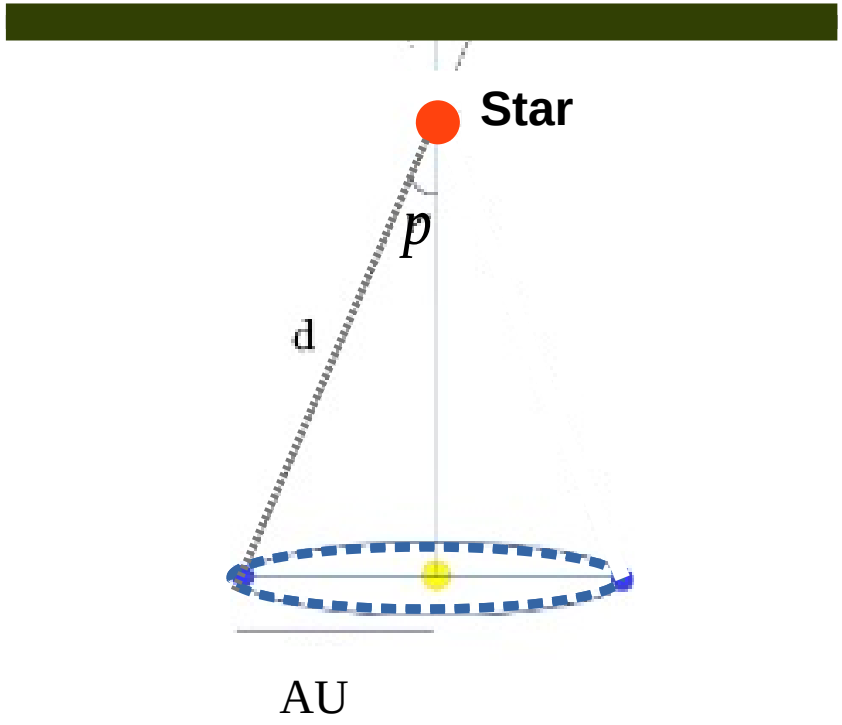


in filter B, V, ...

Definition: colour B-V; V-I ….


distance
temperature
magnitude
size
Object properties

Distance to stars and stellar parallax

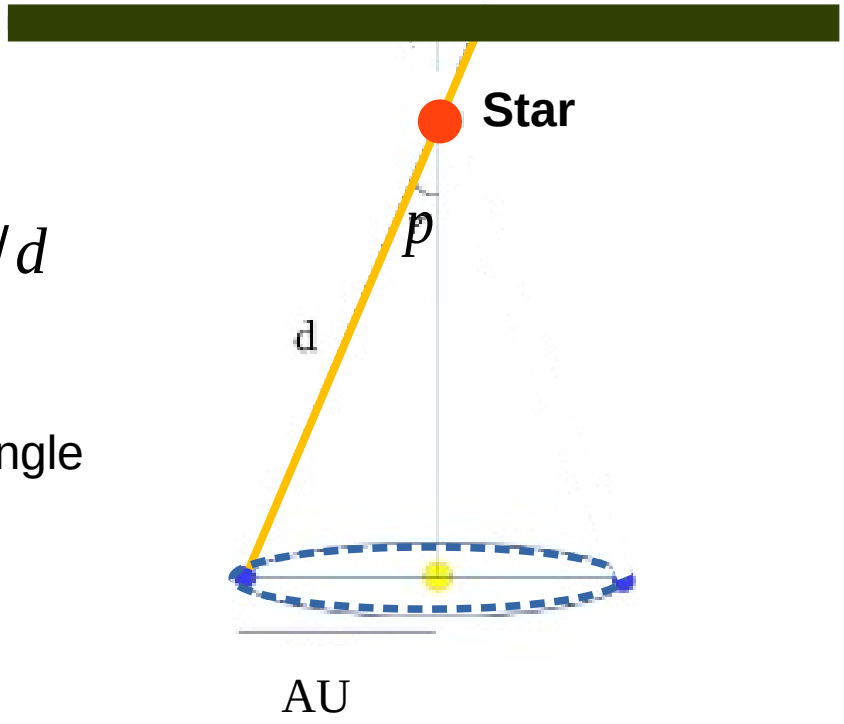


http://pt.wikipedia.org/wiki/Paralaxe

Distance to stars and stellar parallax

$$\sin p = 1\,AU / d$$

p - Parallax angle

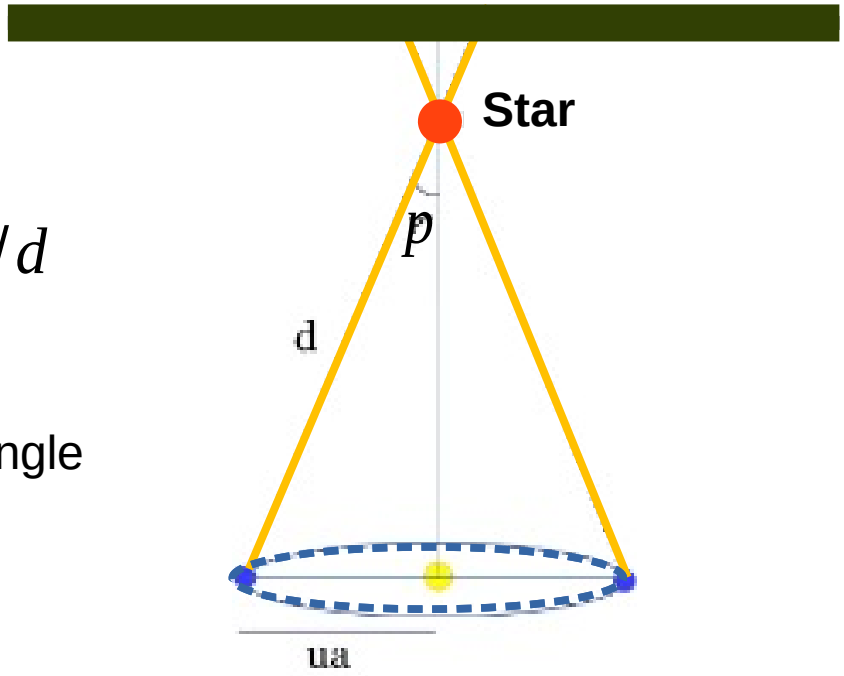


http://pt.wikipedia.org/wiki/Paralaxe

Distance to stars and stellar parallax

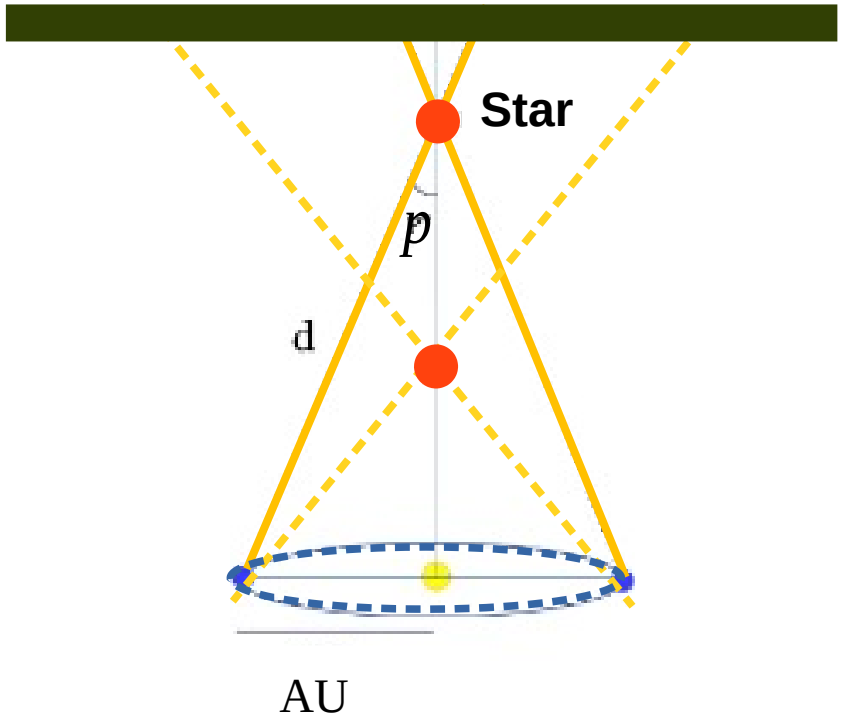
$$\sin p = 1\,AU / d$$

p - Parallax angle



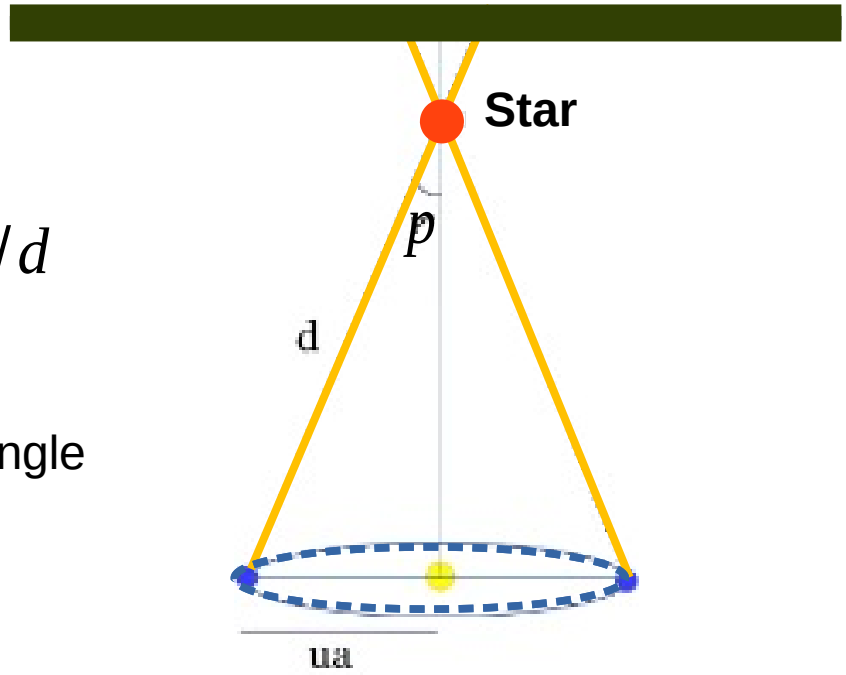
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Distance to stars and stellar parallax



http://pt.wikipedia.org/wiki/Paralaxe

Distance to stars and stellar parallax



$$\sin p = 1\,AU/d$$

p - Parallax angle

$$\sin x = x$$

Radians to degrees

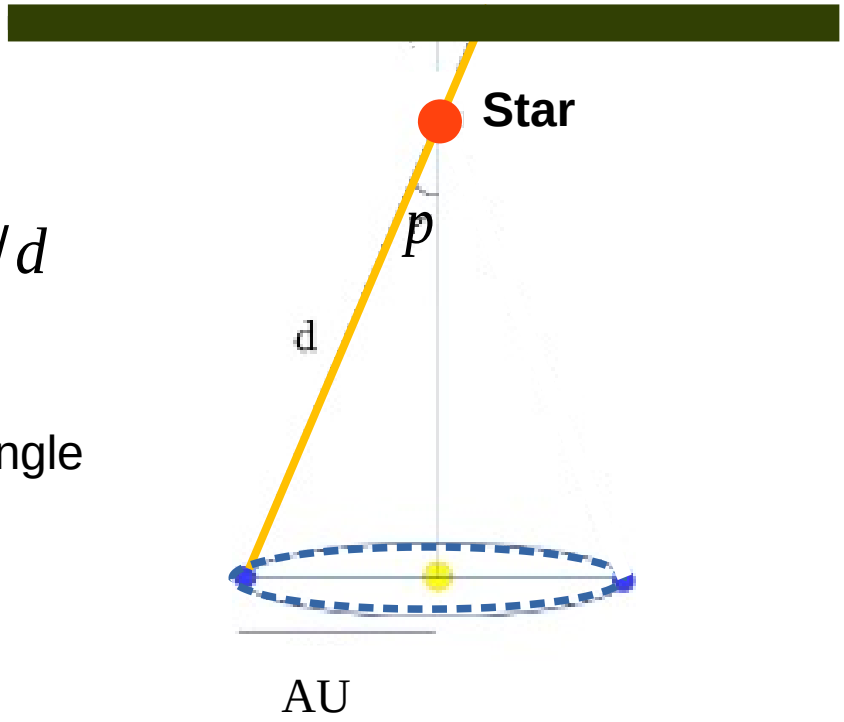
$$x\left(180/\pi\right)graus = x\left(180\,x\,3600/\pi\right)arcsec$$

$$\sin p = p = 1\,AU/d\left(180/\pi\right)graus = 1\,AU/d\left(180\,x\,3600/\pi\right)arcsec$$

http://pt.wikipedia.org/wiki/Paralaxe

Distance to stars and stellar parallax

parsec



$\sin p = 1\,AU / d$

p - Parallax angle

When $p = 1''$

$$1'' = 1\,AU / d \Rightarrow d = 1/1''\,AU = \left(180\,x\,3600/\pi\right) AU = 1\,pc$$

$$d = 206264\,AU = 3{,}2\,ly = 1\,pc$$

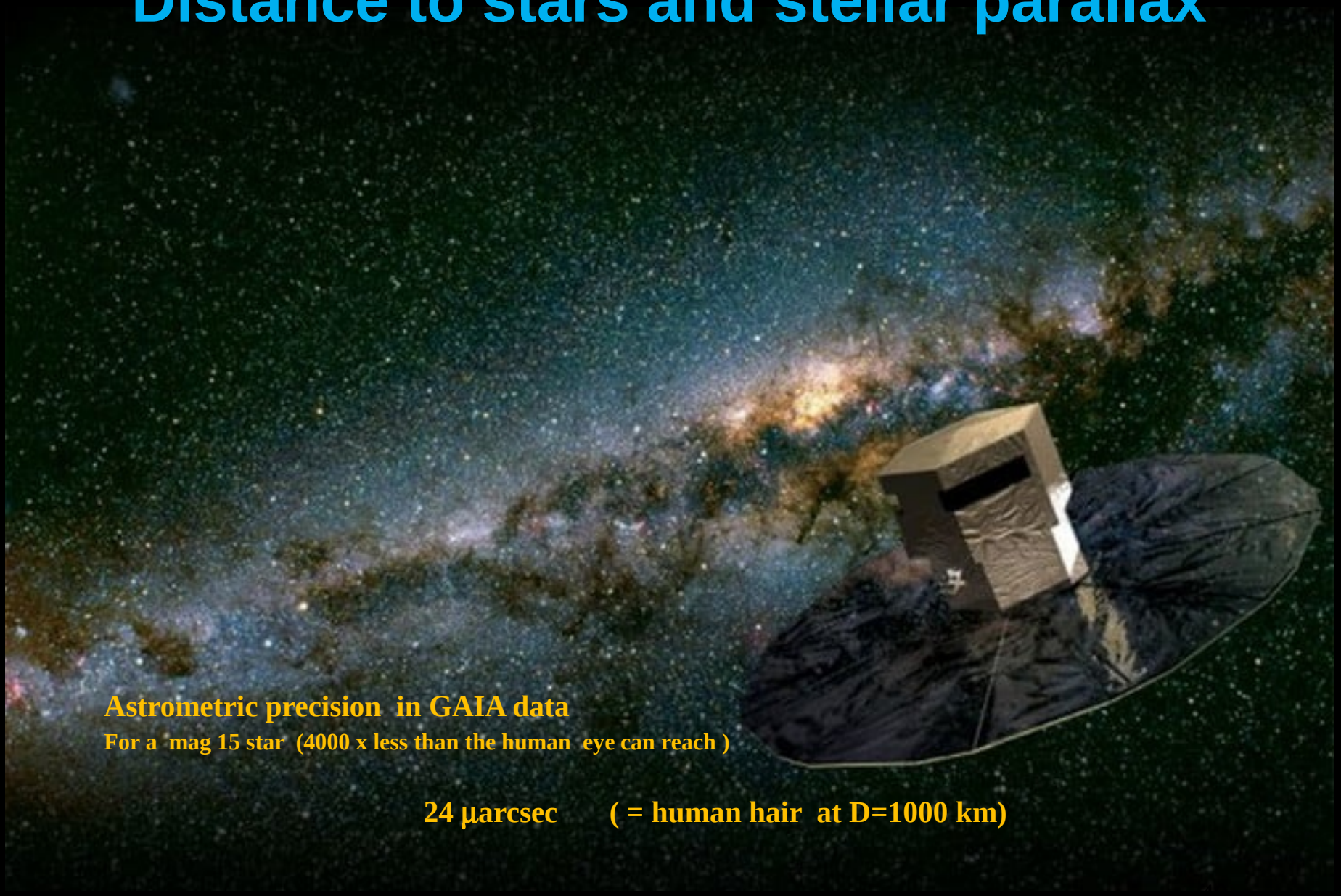


M87, Hubble telescope (ESA,NASA)

D=16,4 Mpc

jet extends to 1500 pc

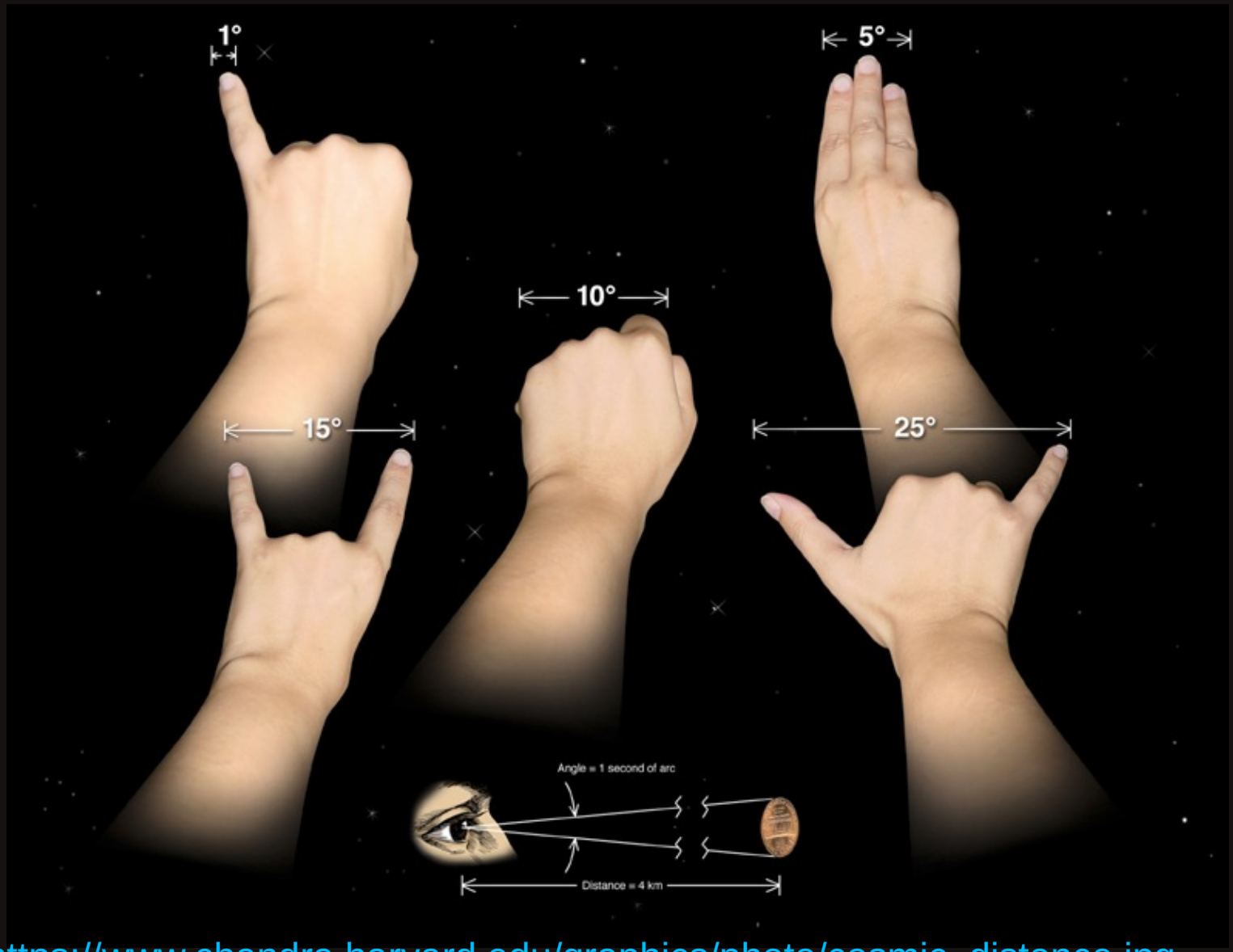
Distance to stars and stellar parallax



Astrometric precision in GAIA data

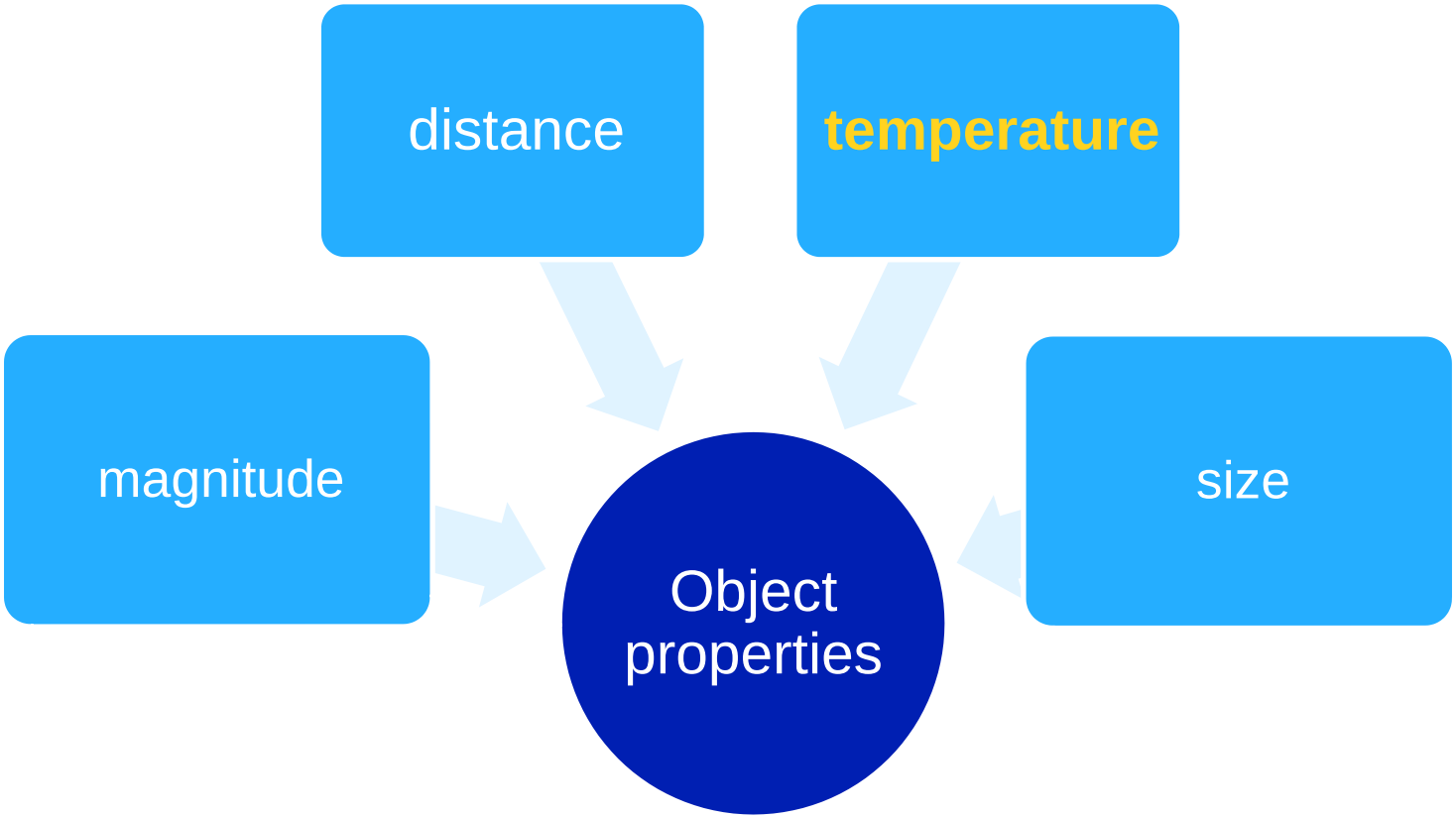
For a mag 15 star (4000 x less than the human eye can reach)

24 μarcsec (= human hair at D=1000 km)



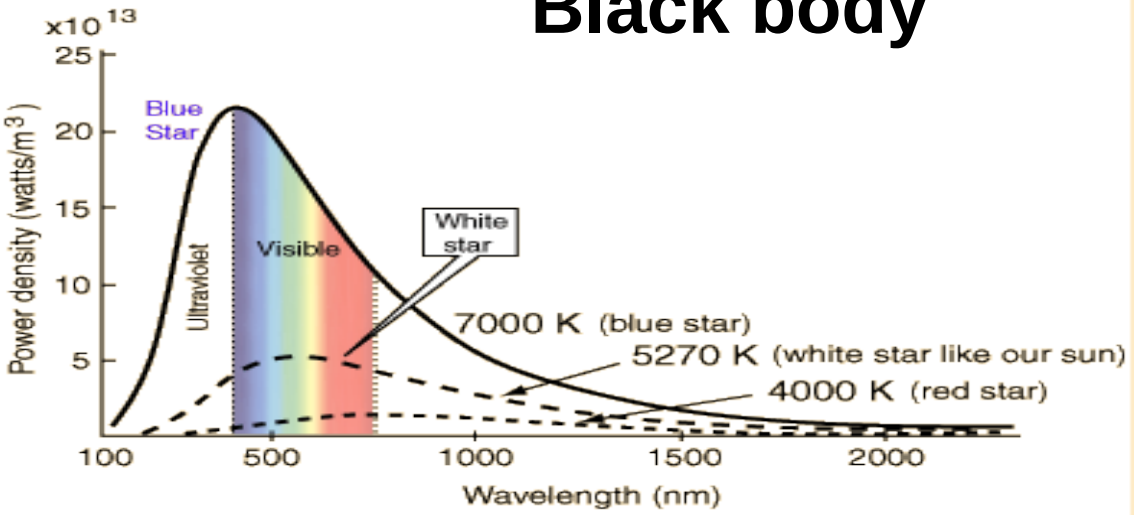
https://www.chandra.harvard.edu/graphics/photo/cosmic_distance.jpg

PROPERTIES OF THE TARGETS



Stellar temperature

Black body





Planck

$$L = 4\pi R^2 \sigma T_{eff}^4$$

R- radius

σ-Stefan Boltzmann constant

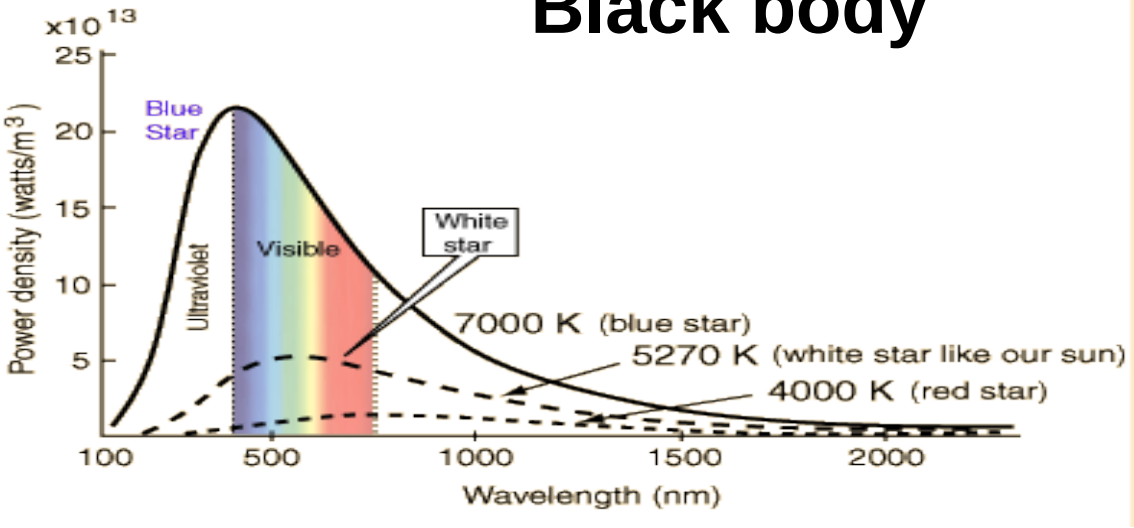
T_{eff} – effective temperature: temperature of blackbody with same integrated surface flux as star

$$\lambda_{max} T = 0{,}28\ cm\,K$$

Wien's Law

Stellar temperature

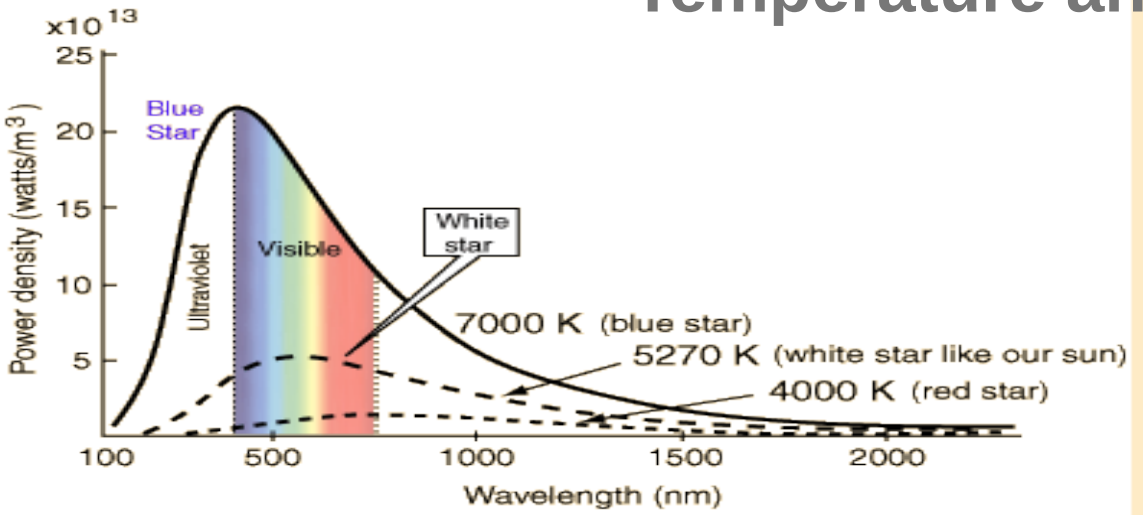
Black body



How can we estimate T ?

Stellar temperature

Temperature and Colour



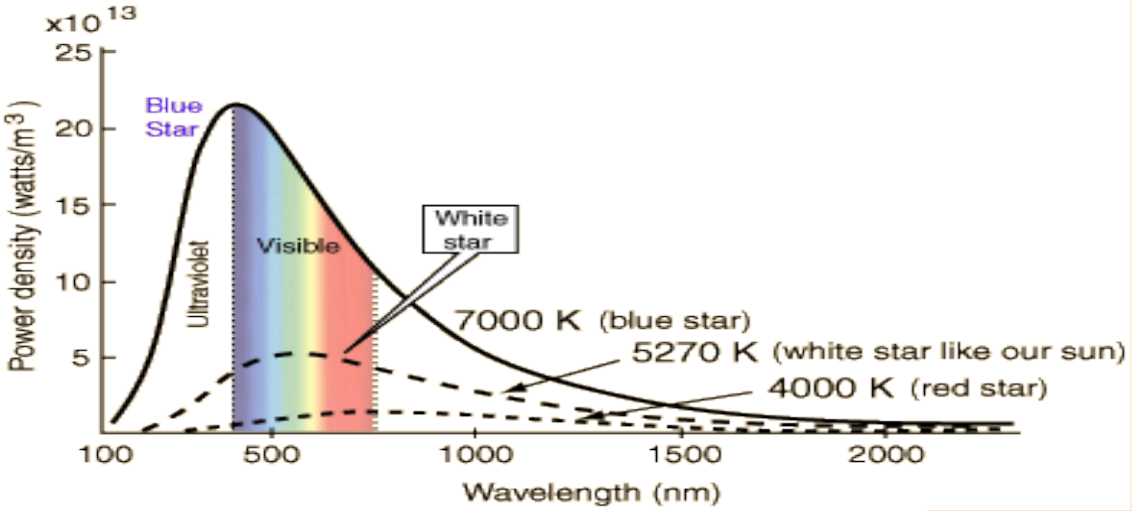
$$mag\,B - mag\,V = -2{,}5\left(\log F_B - \log F_V\right) = -2.5\log\left(L_B / L_V\right)$$

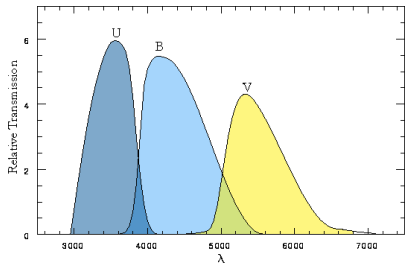
Does not depend on the distance

Depends on the temperature

Stellar temperature

Temperature and Colour





Color index

$U-B = m_U - m_B$

$B-V = m_B - m_V$

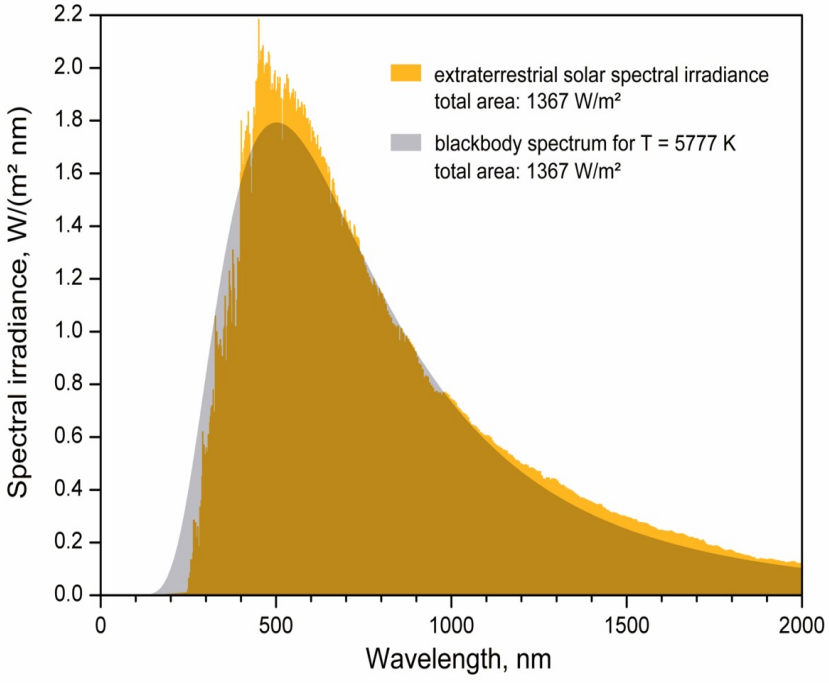
A star with a smaller (B-V) color index is bluer than the one with higher (B-V)

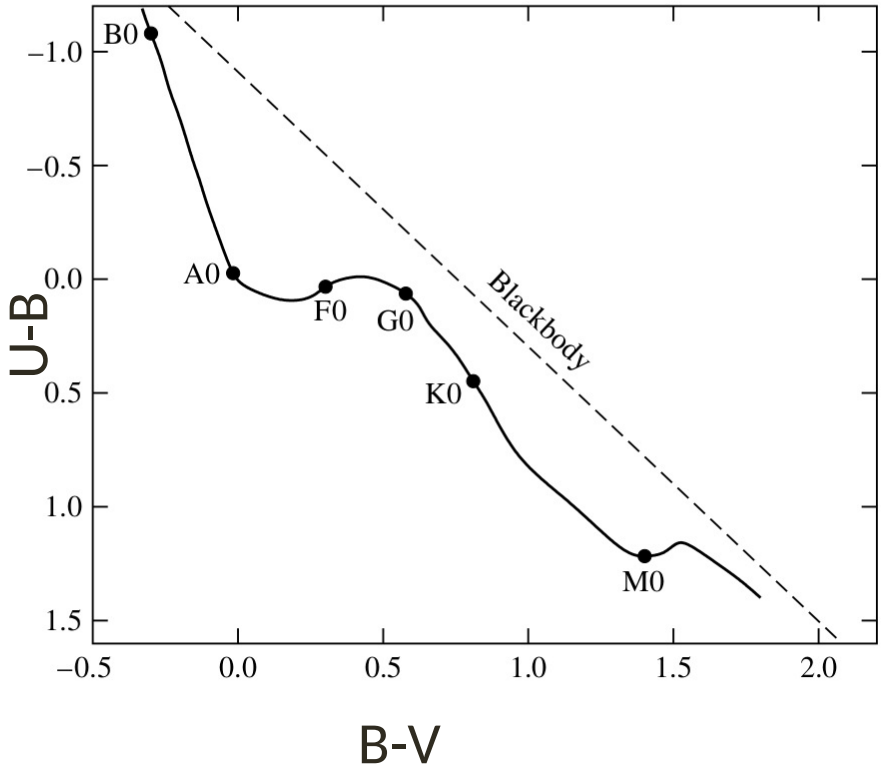
Typical optical filters: UBVRI, ugriz

- In practice, astronomers observe
- with different **passbands** or **filters**

Stellar temperature

Temperature and Colour





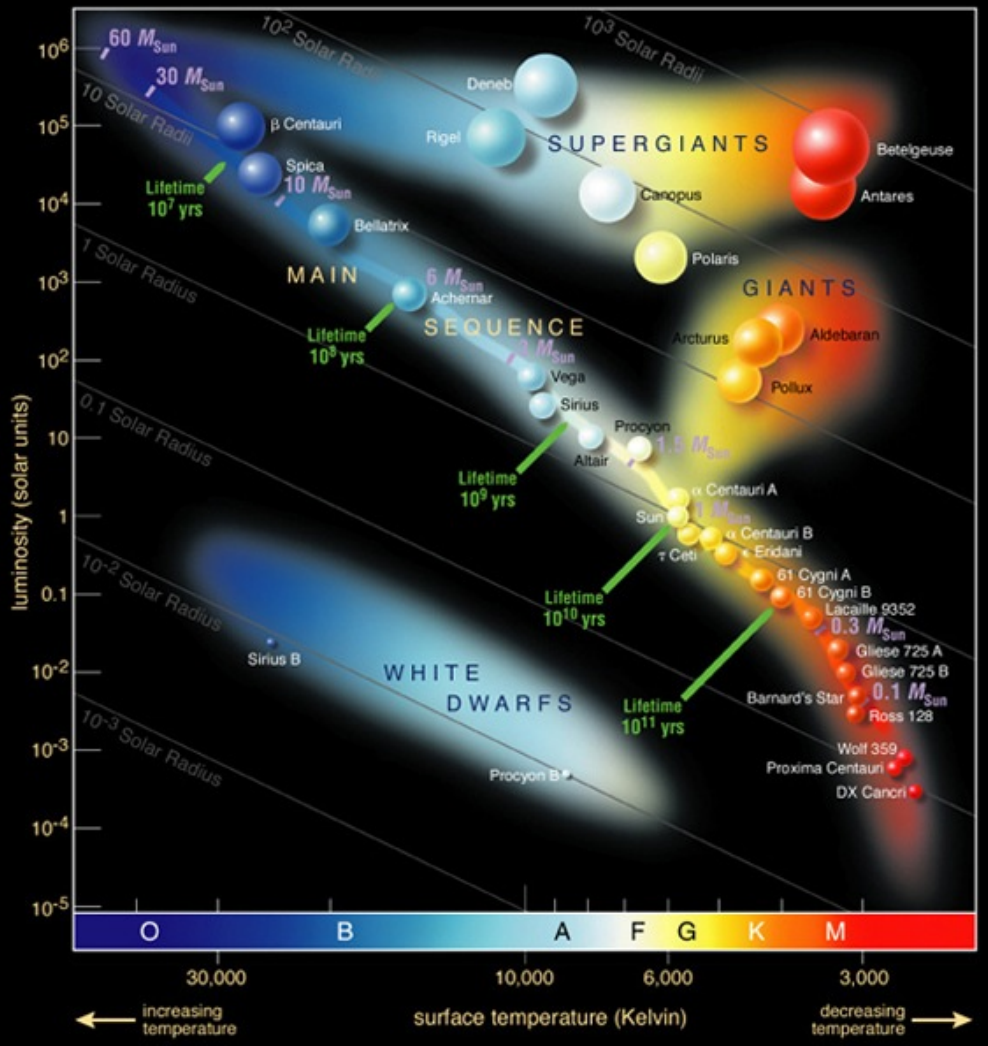
O	**30,000 - 60,000 K**	**Blue stars**
B	**10,000 - 30,000 K**	**Blue-white stars**
A	**7,500 - 10,000 K**	**White stars**
F	**6,000 - 7,500 K**	**Yellow-white stars**
G	**5,000 - 6,000 K**	**Yellow stars (like the Sun)**
K	**3,500 - 5,000K**	**Yellow-orange stars**
M	**< 3,500 K**	**Red stars**

Spectral Class Characteristics **Anie Canon, 1901**

Data from J. C. Evans, George Mason University

Spectral Class	Intrinsic Color	Temperature (K)	Prominent Absorption Lines
O	Blue	41,000	He+, O++, N++, Si++, He, H
B	Blue	31,000	He, H, O+, C+, N+, Si+
A	Blue-white	9,500	H(strongest), Ca+, Mg+, Fe+
F	White	7,240	H(weaker), Ca+, ionized metals
G	Yellow-white	5,920	H(weaker), Ca+, ionized & neutral metal
K	Orange	5,300	Ca+(strongest), neutral metals strong, H(weak)
M	Red	3,850	Strong neutral atoms, TiO

Diagram H-R



Deneb

Arcturus

Aldebaran

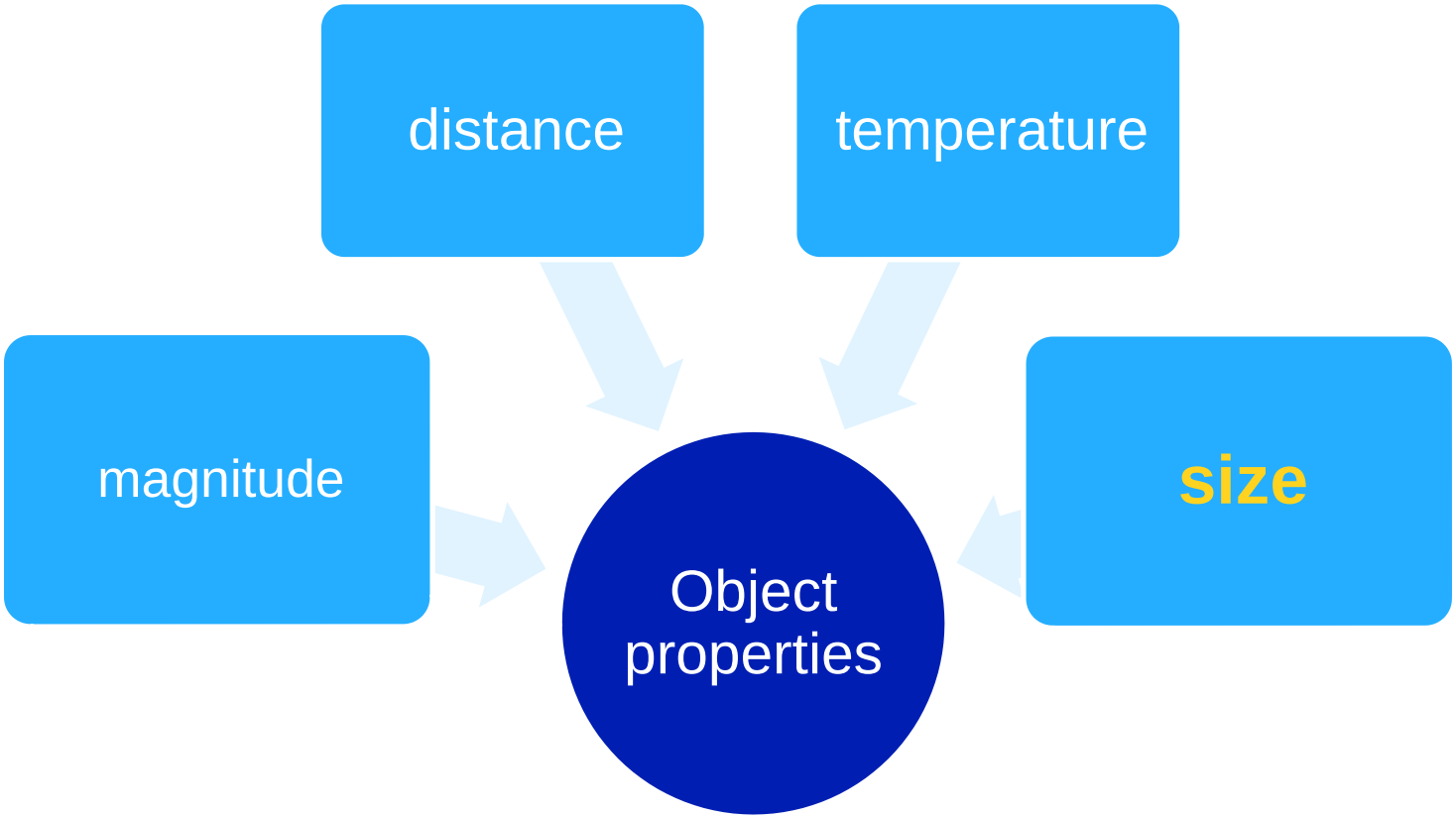
Vega

Altair

Sun

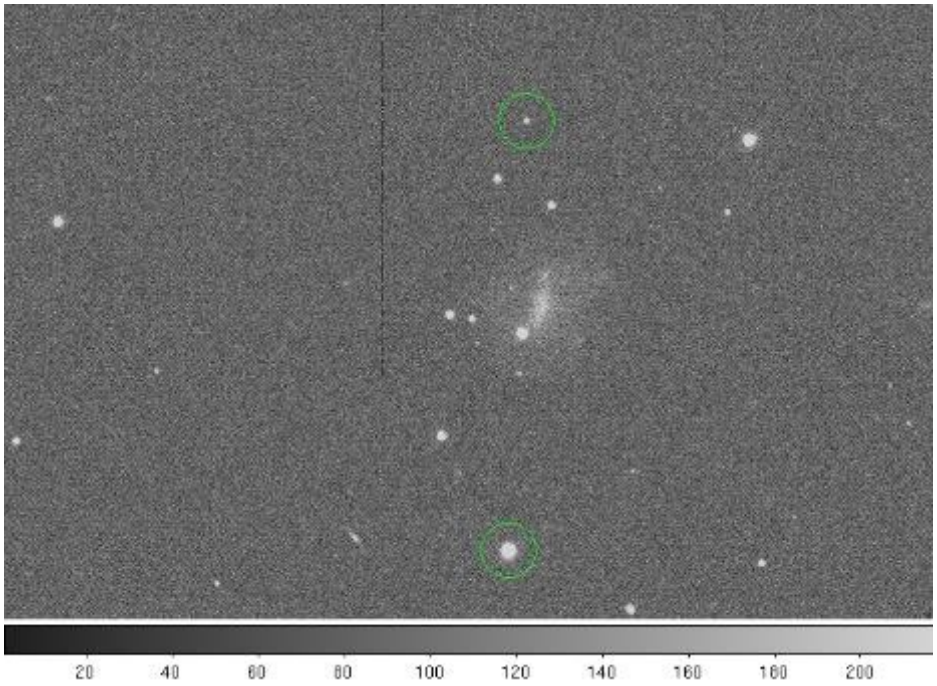
Sirius B

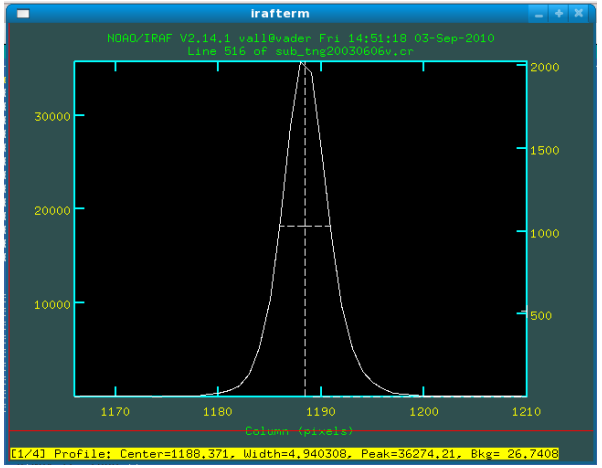
PROPERTIES OF THE TARGETS

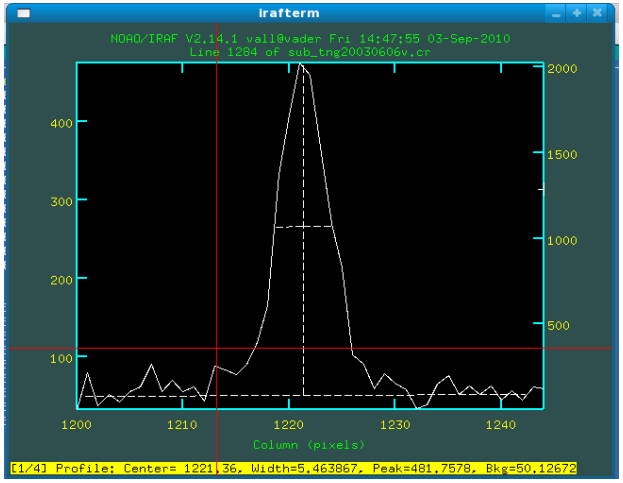


PROPERTIES OF THE TARGETS: size

- Point sources: which star is bigger?
- (the one up or down)?

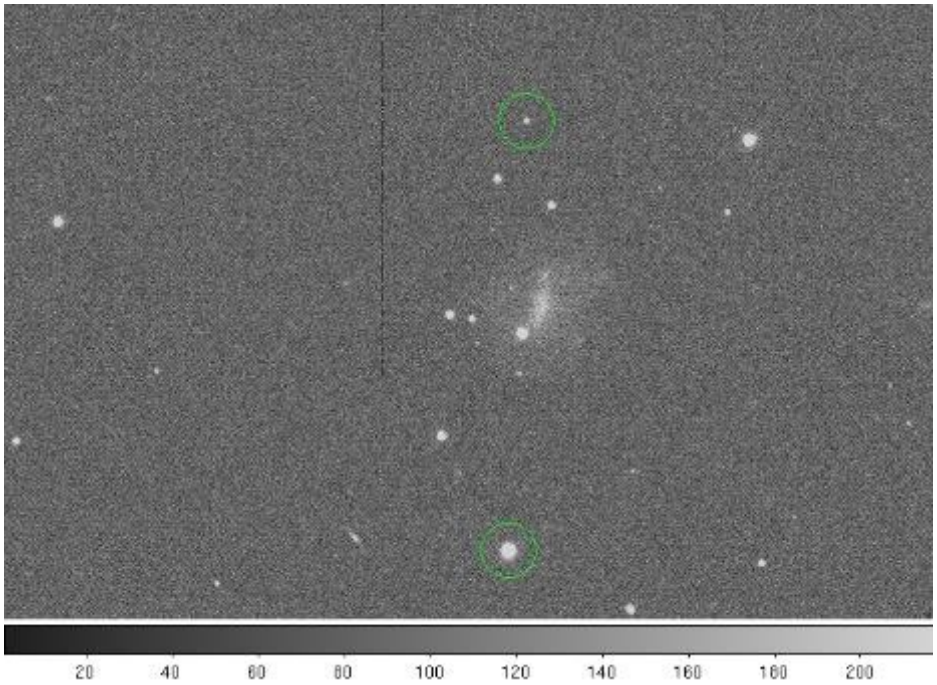


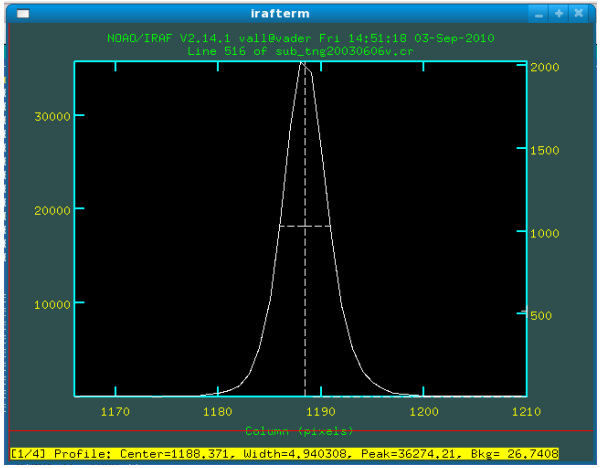


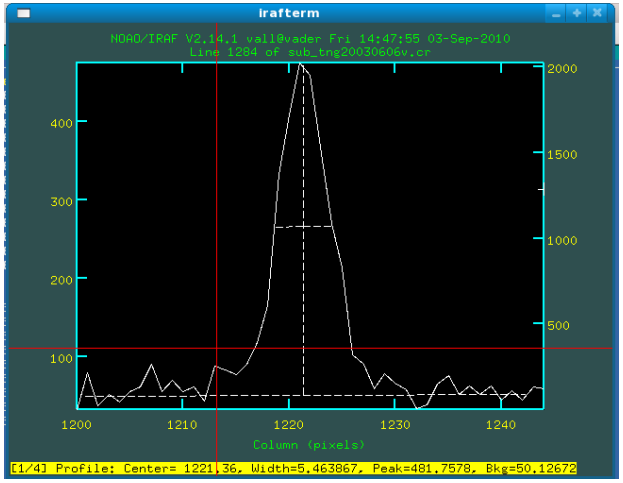


PROPERTIES OF THE TARGETS: size

- All point sources are of equal
- angular sizes (equal fwhm)



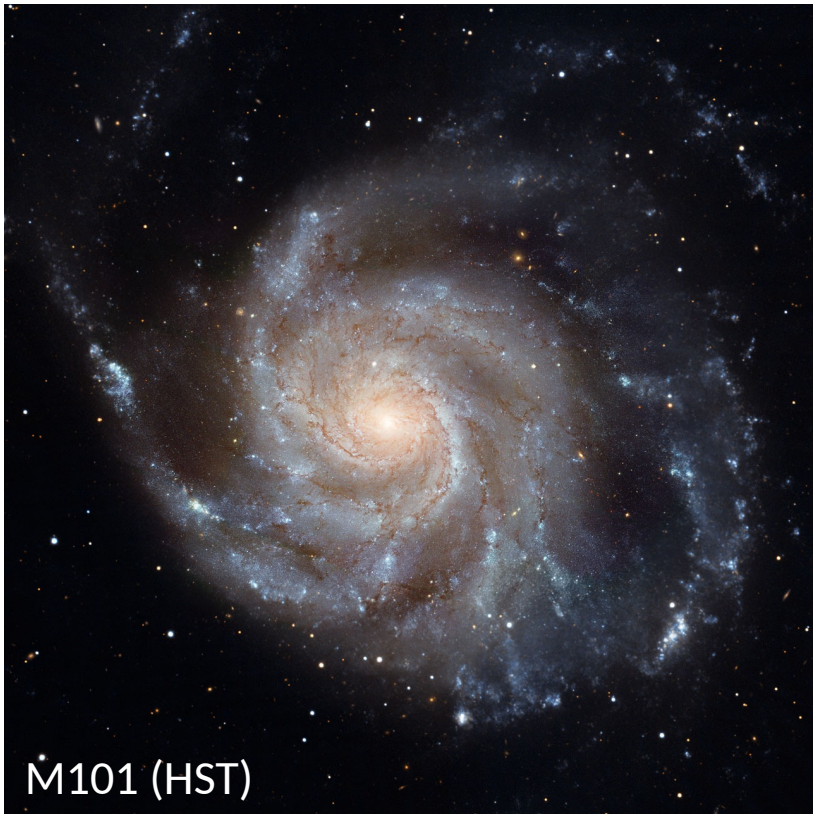




Important notions:
seeing and diffraction limit

PROPERTIES OF THE TARGETS

- Extended sources: surface brightness: mag/arcsec^2



V-band surface brightness:

- Orion:
  17 mag/arcsec^2
- Andromeda:
  14-18.0 mag/arcsec^2
- M101:
  13 mag/arcsec^2

TÉCNICO LISBOA

ASTROPHYSICS LABORATORY
Observational Astrophysics

II) Basic notions in Observational astrophysics

a) Properties of the astronomical objects

- Magnitude; Absolute magnitude
- Magnitude and distance: ly, parsec
- Temperature and colour
- Size

b) Preparing astronomical observations

- Visibility of the target: coordinates and time
- Telescope
- Detector and data acquistion

III) Observations: data acquisition

IV) Data reduction and analysis

OBSERVATIONS

Coordinates of the observer	Latitude and longitude of the observatory
Date and time	Year, day, hour
Coordinates of the targets	Celestial coordinate system
Exposure time	


WHY DO WE NEED COORDINATES
Of the observer on Earth
Of the target in the sky

Earth rotation



Earth rotates along its N/S axis to the E
Stars rise from E to W The center of the circles marks the celestial pole



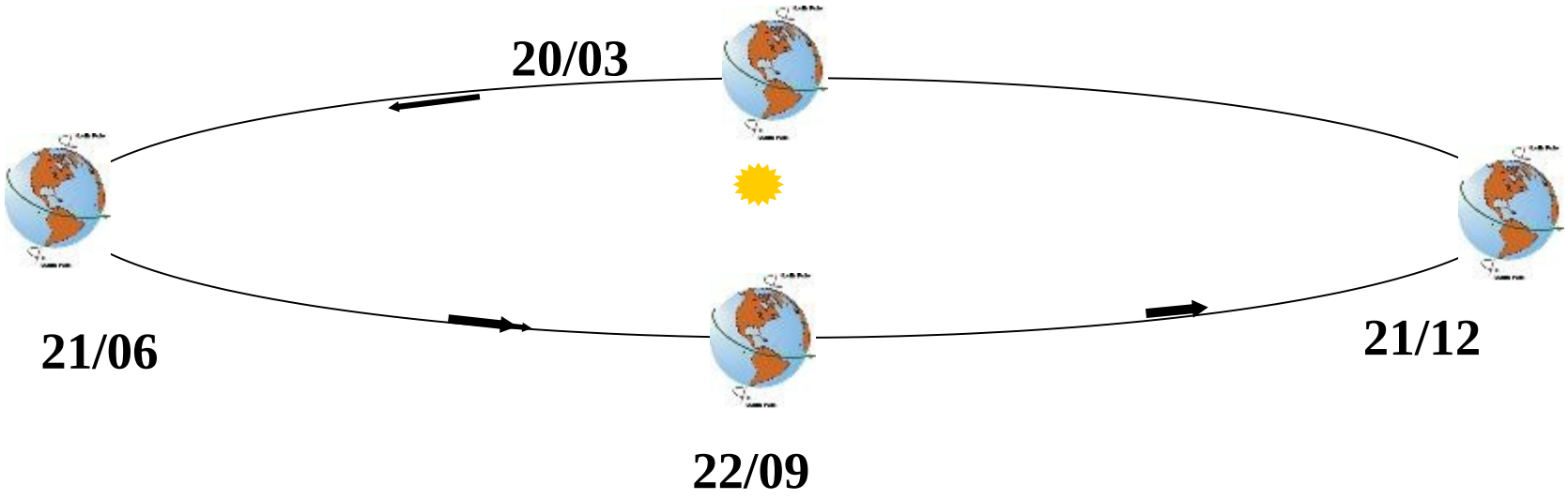
Coordenadas GPS

Latitude: N 38º 46’ 42.03" (38.778342º)

Longitude: W 9º 06’ 49.53" (-9.113758º)

Celestron 14” in Lisbon

IS OUR TARGET VISIBLE?



Where is my target

on a given date?

on a given hour?

Celestial coordinate systems: next lecture